VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (Engels_2022_Cleaner_Production)
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Title: Life cycle assessment of natural graphite production for lithium-ion battery anodes based on industrial primary data
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Sample: N = Primary industrial data from Chinese vertically integrated graphite producer (2019 baseline)

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Mining: 9.59 t ore/t conc; Elec: 0.027 mmBtu/t (0.008 kWh/kg ore); Diesel: 0.082 mmBtu/t (0.024 kWh/kg ore)
• Flotation: Concentrate 85-98% C; Elec: 1.561 mmBtu/t (0.457 kWh/kg); Coal: 1.132 mmBtu/t (1.194 MJ/kg)
• Spheronization: 2.22 t conc/t spherical graphite (45% yield, 55% fines); Elec: 6.478 mmBtu/t (1.899 kWh/kg)
• Chemical Purification: 1.13 t SG/t SPG (>99.95% C); HF: 180 kg/t; HCl: 200 kg/t; HNO3: 100 kg/t; Lime: 400 kg/t; Elec: 0.941 mmBtu/t (0.276 kWh/kg); NG: 0.905 mmBtu/t
• Pitch Coating: 1.01 t SPG + 50 kg HSP pitch/t CSPG (1000-1200C); Elec: 14.034 mmBtu/t (4.113 kWh/kg CSPG)
• Cumulative direct CSPG electricity: 7.91 kWh/kg (28.48 MJ/kg); GWP: 9.62 kg CO2-eq/kg CSPG (Chinese coal grid)

VERBATIM SOURCE QUOTE / TABLE RECORD:

Graphite ore (carbon content: 11%) is mined primarily via open cast mining... Flotation is conducted using grinding mills, water and pine oil as flotation agents, and diesel as a collector... Spheronization: these flakes pass through multiple classifier mills (arranged in a row) to get spherical graphite particles with three desired properties: narrow particle size distribution, high tap density, and low specific surface area. A key byproduct of this step is graphite fines, that constitute 60% of total output from this process... Purification is done by chemically leaching spherical graphite with acids (hydrofluoric, hydrochloric, and nitric), and then treating with an alkaline reagent (quicklime) to neutralize the left-over acids... Coating is carbonized by heating the coated particles in a furnace for 15-16 hours in an inert atmosphere.